GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of canopy height per plot (esy4)

MB

2025-09-30

Contents

> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
```

```r
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites_esy4.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
      ),
    fertilized = "f",
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
  mutate(
    esy4 = fct_relevel(esy4, "R", "R22", "R1A"),
    eco.id = factor(eco.id)
    ) %>%
  rename(y = cwm.abu.height) %>%
  filter(y < 1) # see section Outliers: Exclude site N_DAM (more or less only the tall grass Arrhenathe
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##     upper      mean     lower
## 0.4776193 0.4676948 0.4577703
```

```r
median(sites$y)
```

```
## [1] 0.47
```

```r
sd(sites$y)
```

```
## [1] 0.1254282
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##   5%  95%
## 0.27 0.68
```

```r
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
```

```
##   eco.id     n
##   <fct>  <int>
## 1 654      202
## 2 664      199
## 3 686      215
```

```r
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type     n
##   <ord>     <int>
## 1 positive    102
## 2 restored    396
## 3 negative    118
```

```r
sites %>% count(esy4)
```

```
## # A tibble: 3 x 2
##   esy4      n
##   <fct> <int>
## 1 R       327
## 2 R22     208
## 3 R1A      81
```

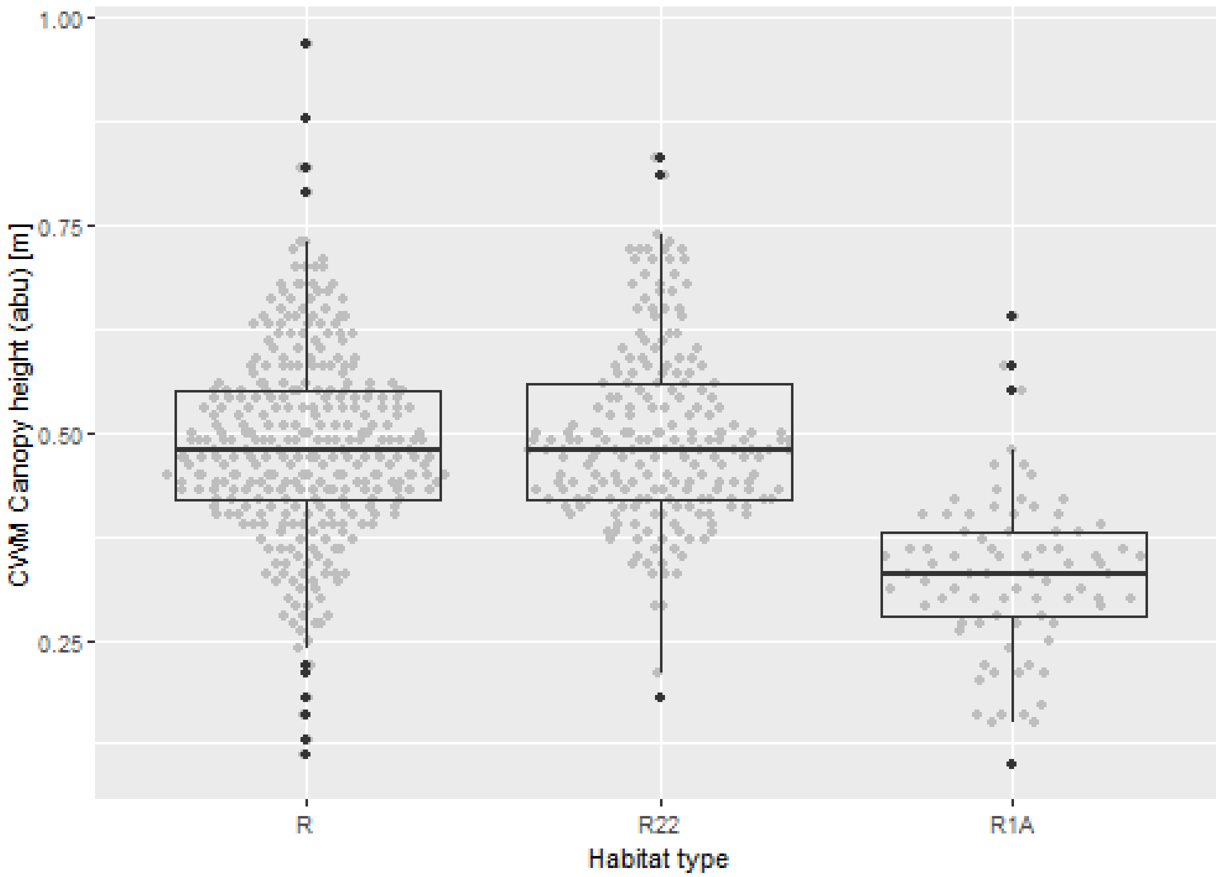
```r
sites %>% count(esy4, eco.id)
```

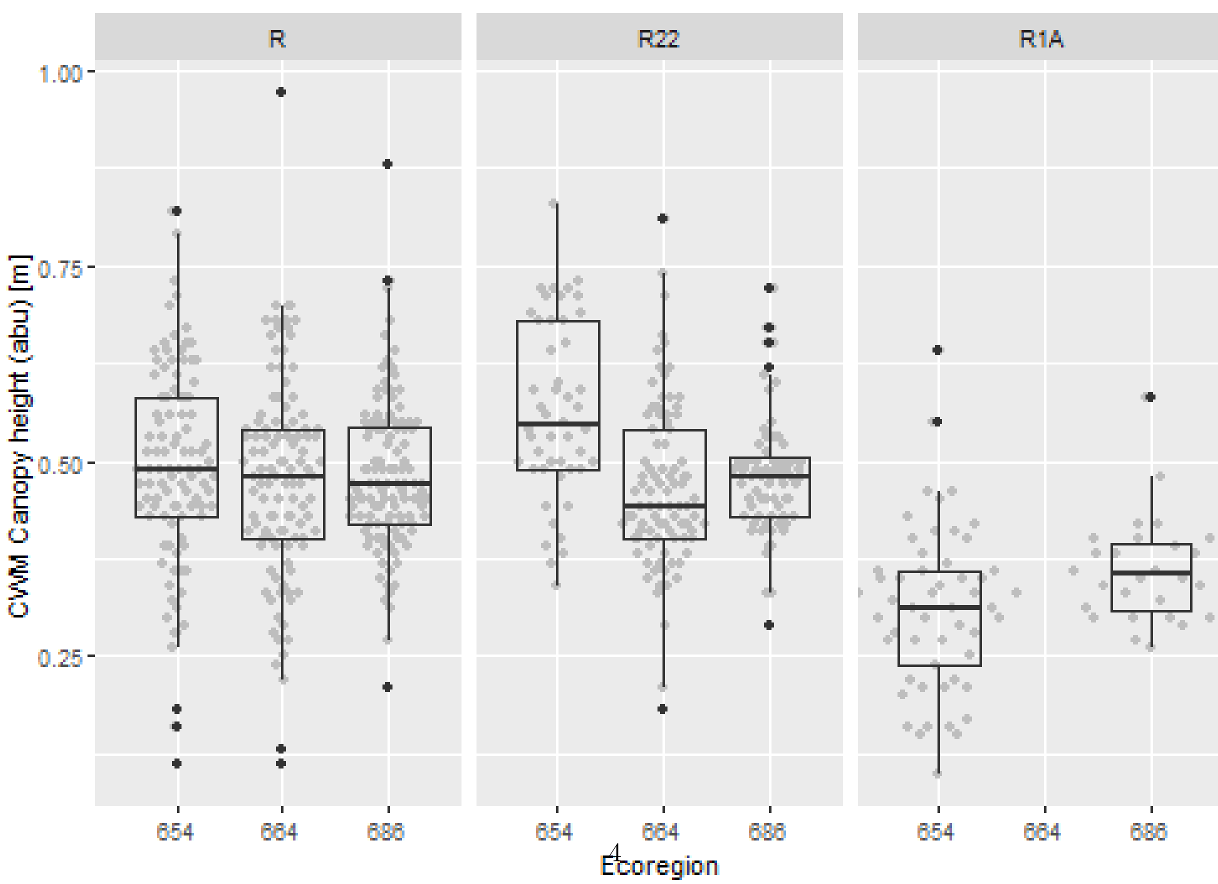
```
## # A tibble: 8 x 3
##   esy4  eco.id     n
##   <fct> <fct>  <int>
## 1 R     654      101
## 2 R     664      110
## 3 R     686      116
## 4 R22   654       48
## 5 R22   664       89
## 6 R22   686       71
## 7 R1A   654       53
## 8 R1A   686       28
```

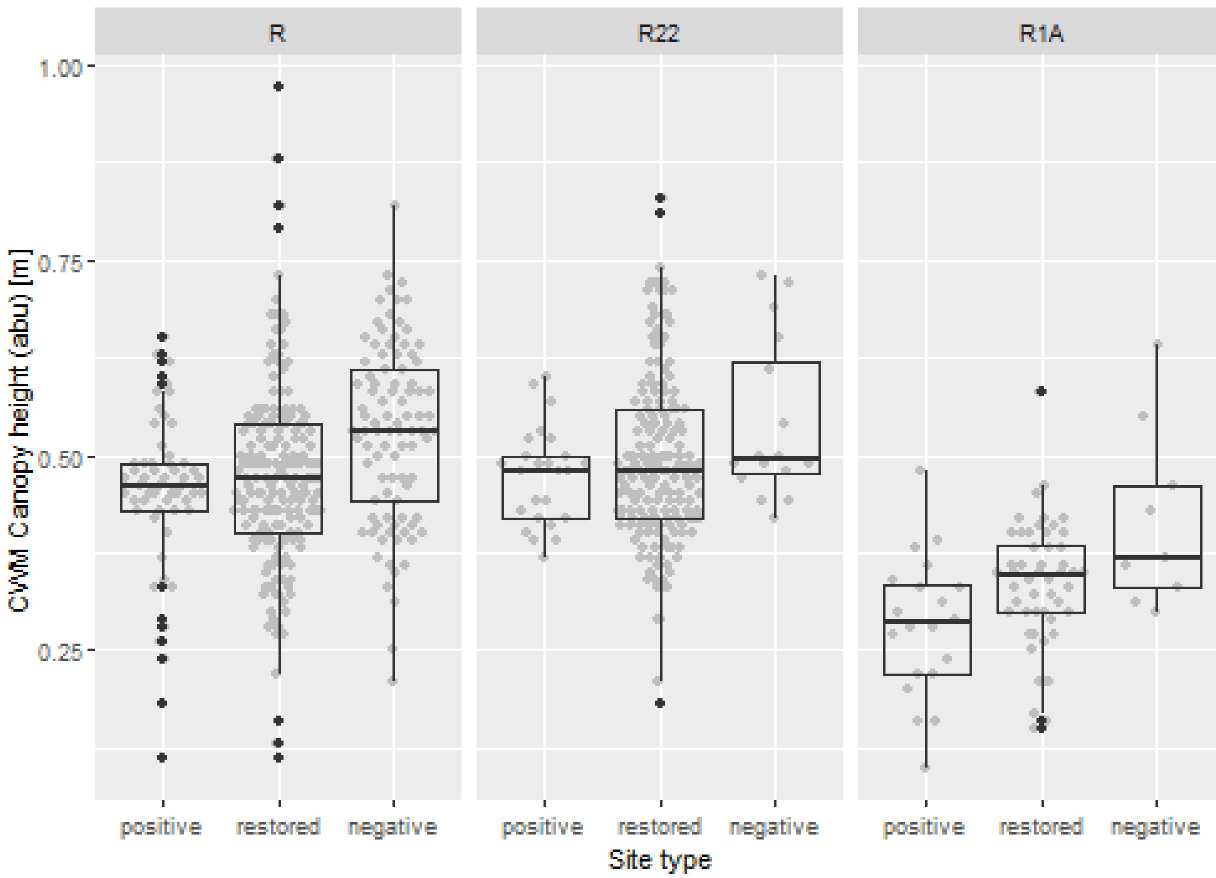
```r
sites %>% count(esy4, site.type)
```

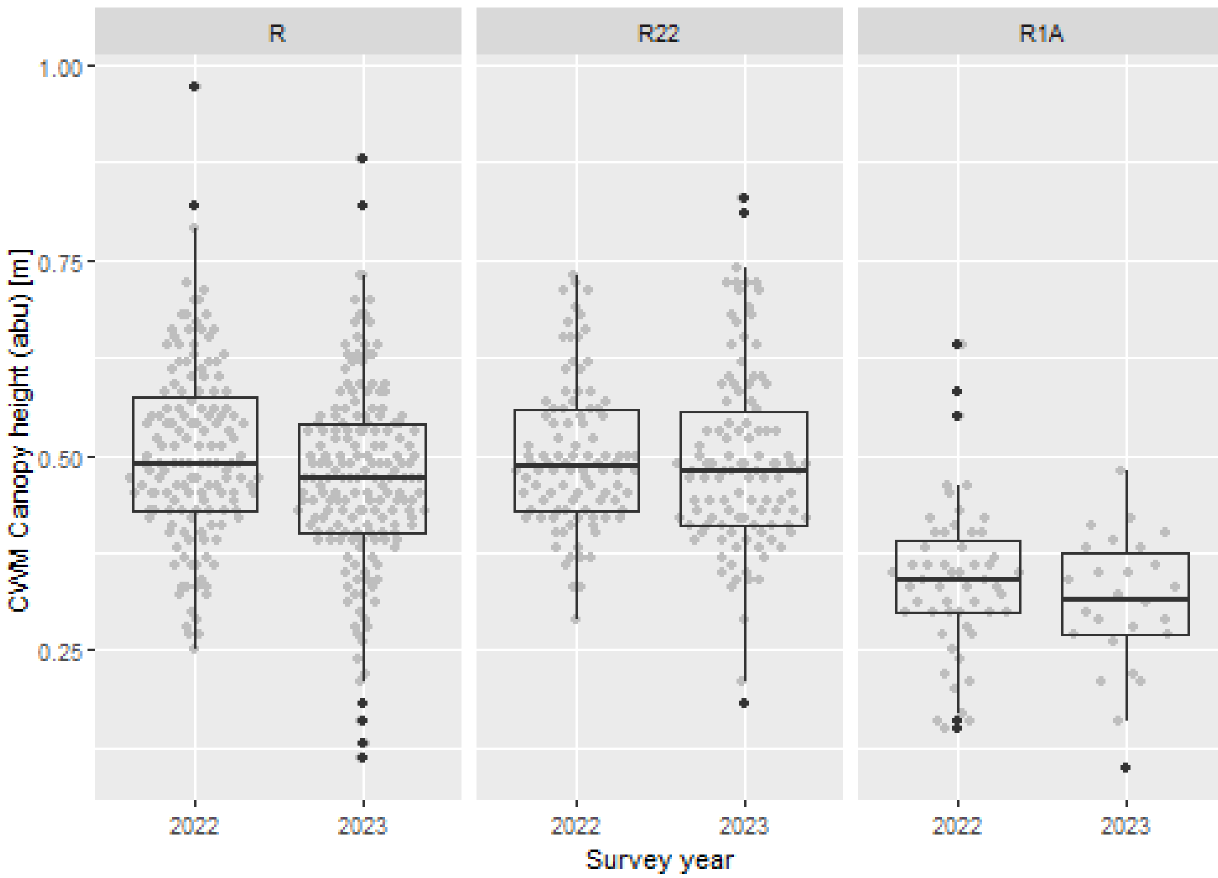
```
## # A tibble: 9 x 3
##   esy4  site.type     n
##   <fct> <ord>     <int>
## 1 R     positive     57
## 2 R     restored    177
## 3 R     negative     93
## 4 R22   positive     25
## 5 R22   restored    167
## 6 R22   negative     16
## 7 R1A   positive     20
## 8 R1A   restored     52
## 9 R1A   negative      9
```

Graphs of raw data (Step 2, 6, 7)

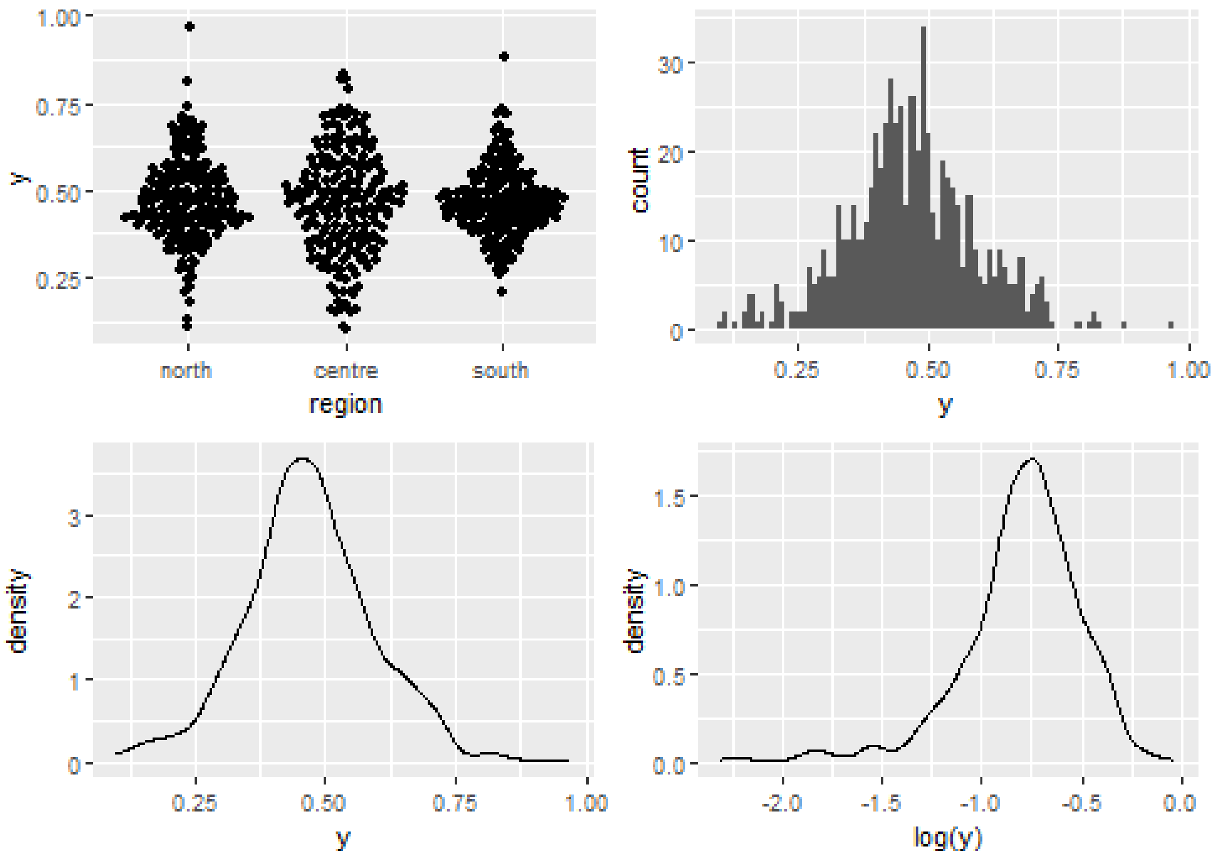





R
R22
R1A
1.00
0.75
0.50
0.25
CWM Canopy height (abu) [m]
positive
restored
negative
positive
restored
negative
positive
restored
negative
Site type


R
R22
R1A
1.00
0.75
0.50
0.25
CWM Canopy height (abu) [m]
2022
2023
2022
2023
2022
2023
Survey year

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_height_esy4_2.Rdata"))
load(file = here("outputs", "models", "model_height_esy4_3.Rdata"))
m_1 <- m2
m_2 <- m3
```

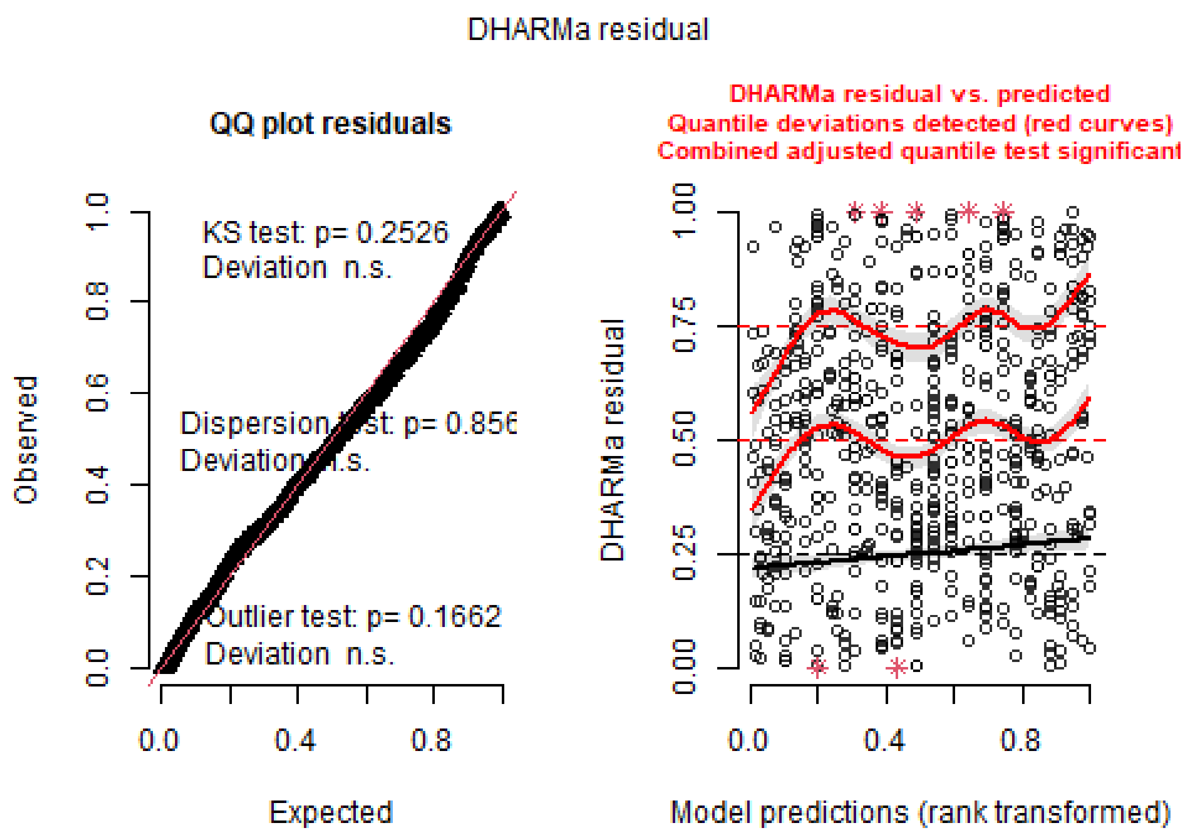
```
m_1@call
## lmer(formula = y ~ esy4 * site.type + eco.id + obs.year + (1 |
##     id.site), data = sites, REML = FALSE)
```

```r
m_2@call
## lmer(formula = y ~ esy4 * (site.type + eco.id) + obs.year + hydrology +
##     (1 | id.site), data = sites, REML = FALSE)
```

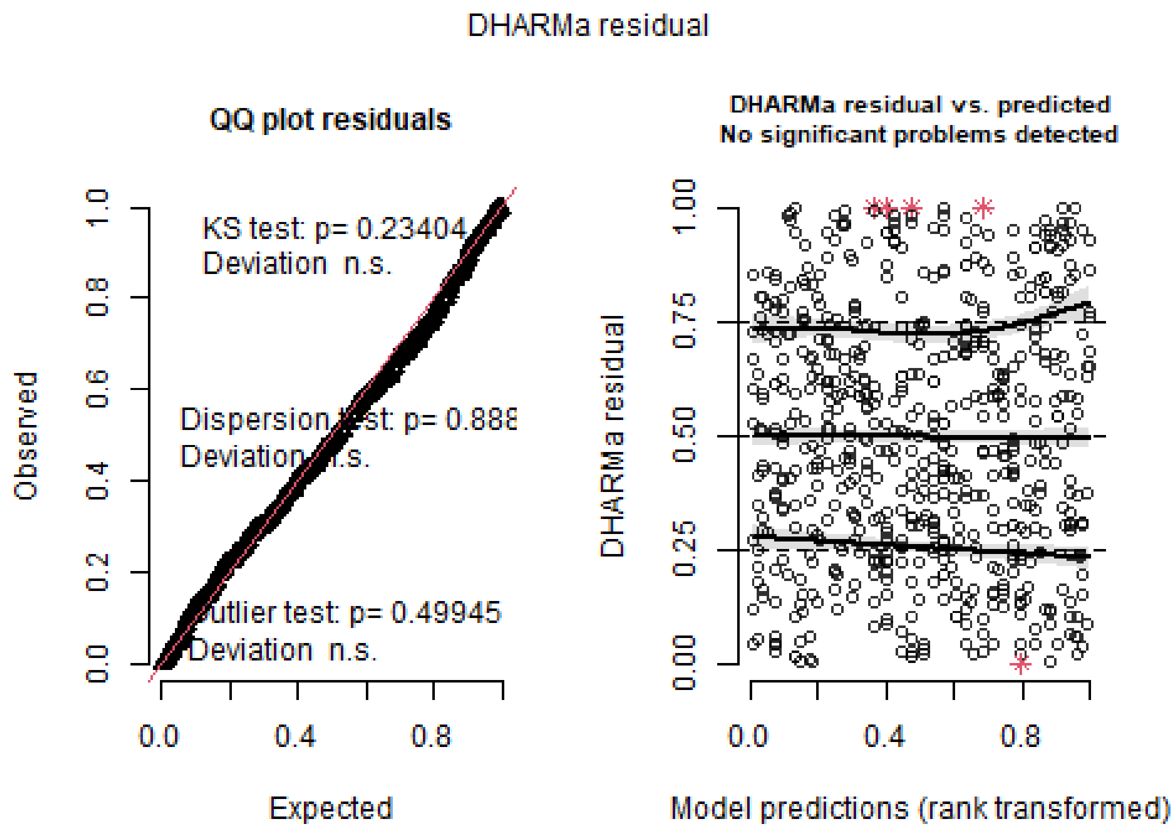
Model check

DHARMa

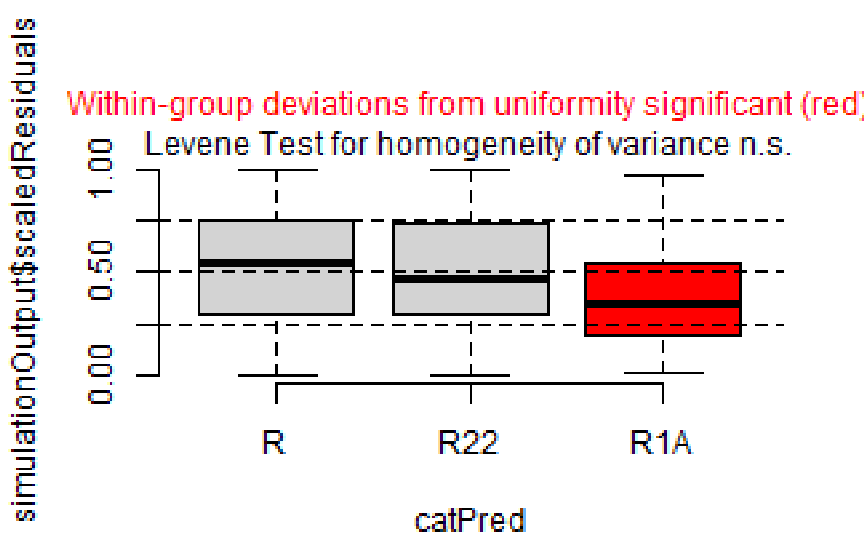
```r
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



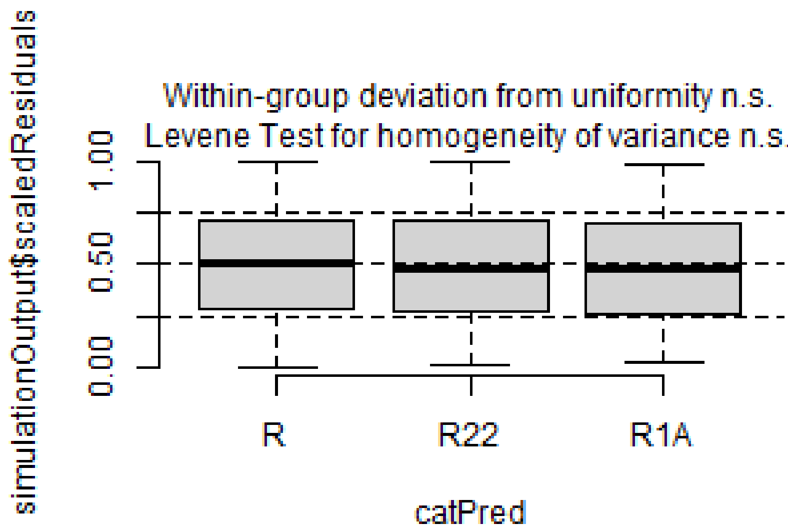

```r
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



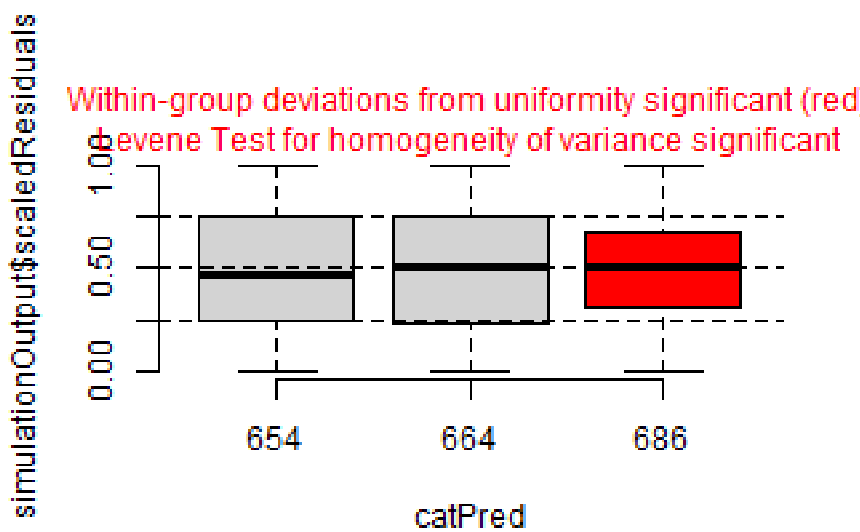

```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



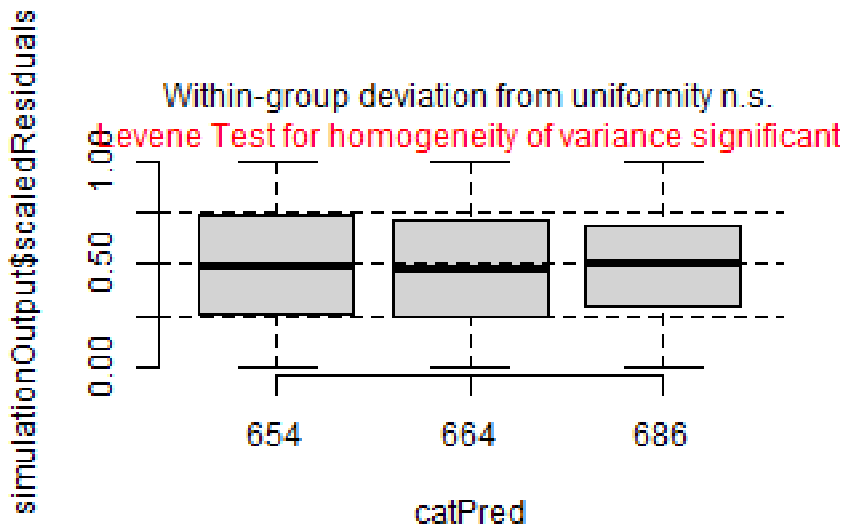

```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



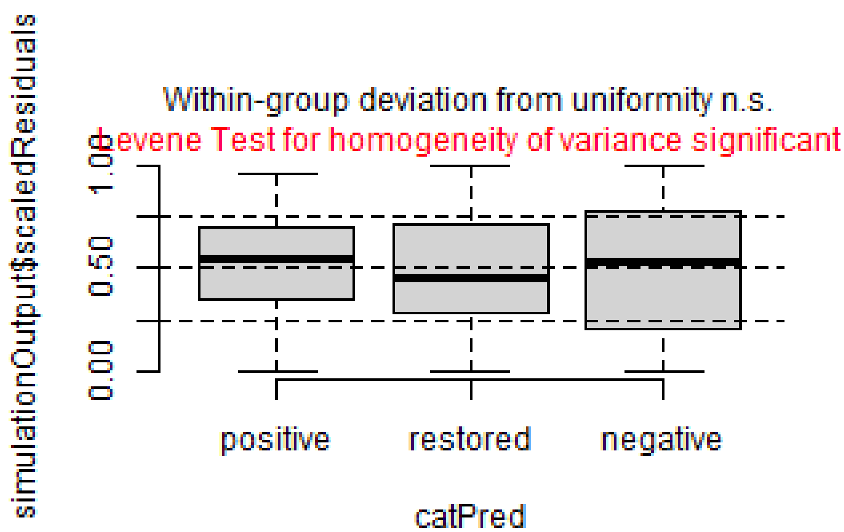

```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



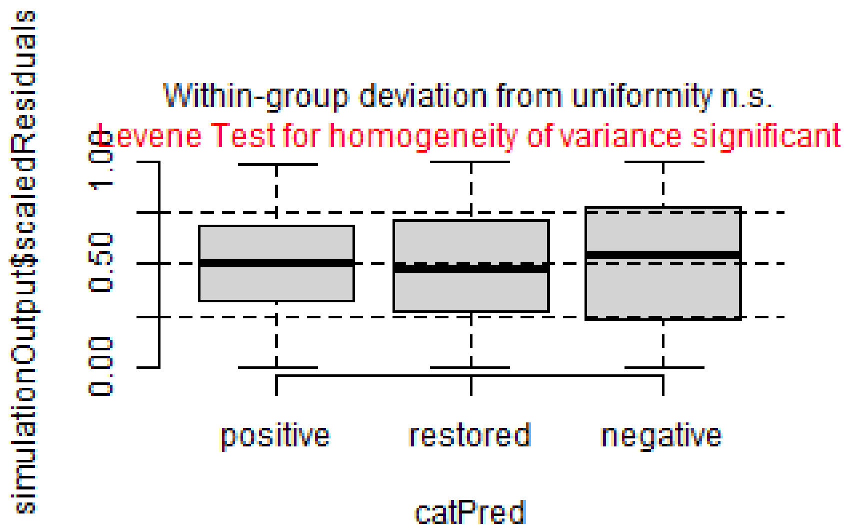

```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



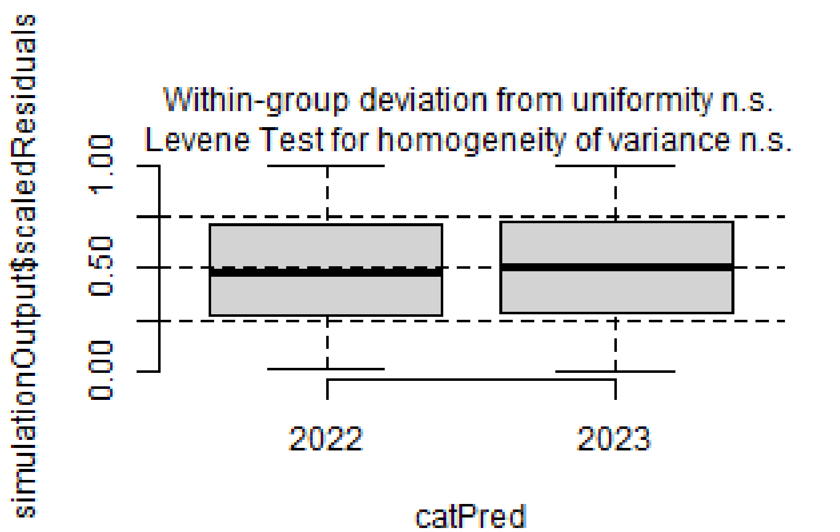

```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



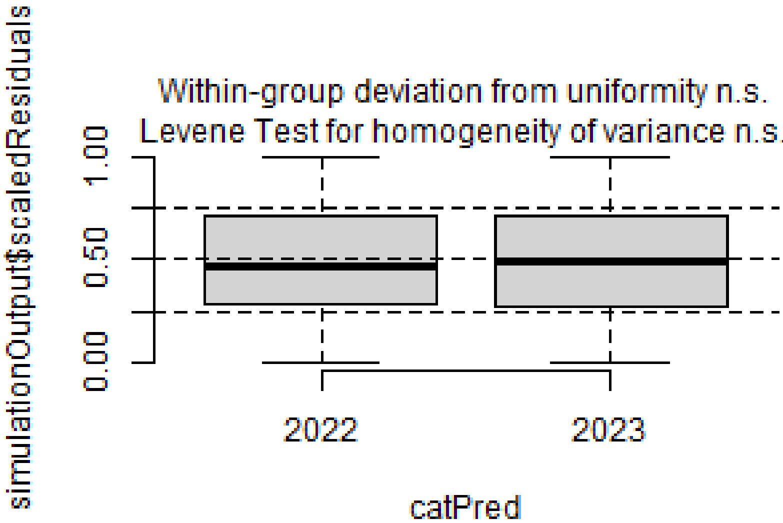

```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



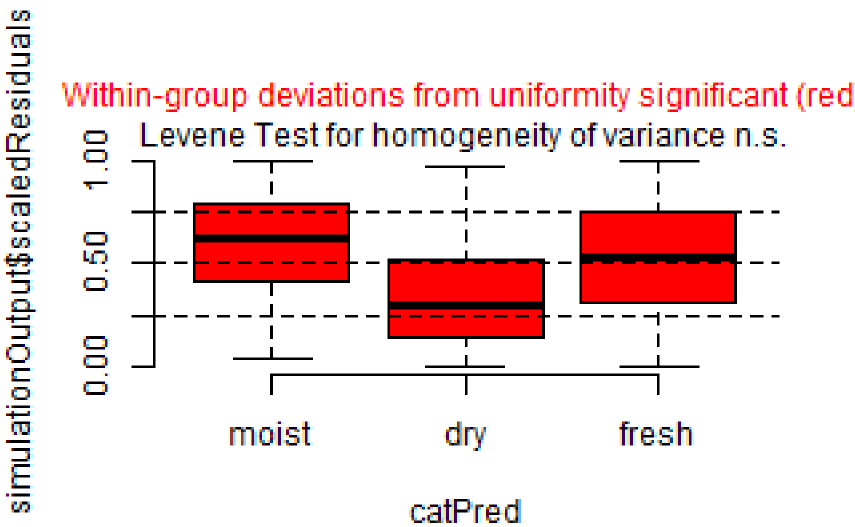

```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



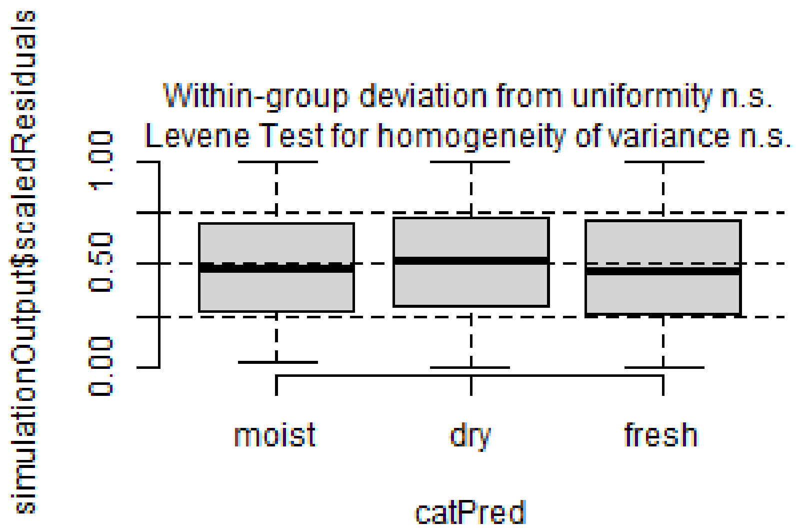

```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



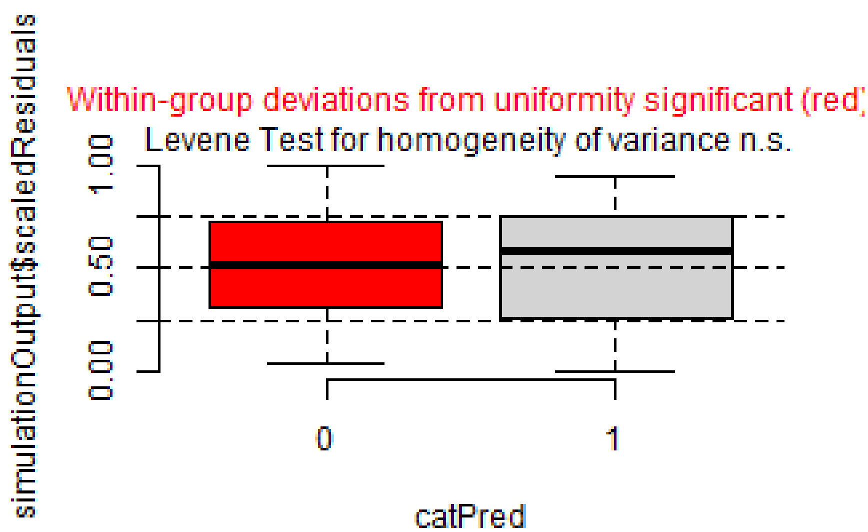

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



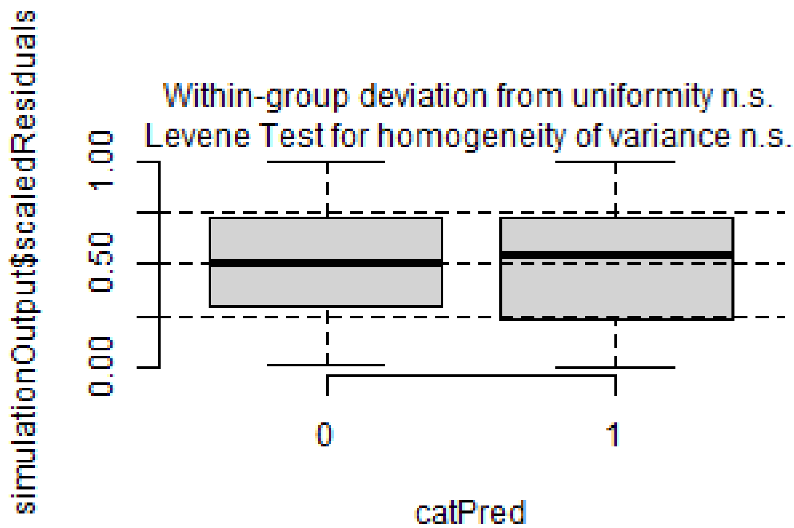

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



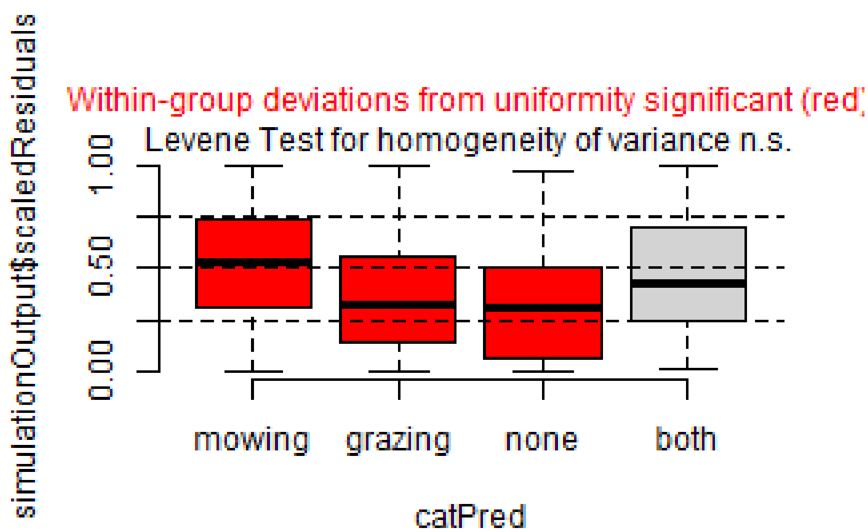

```
plotResiduals(simulation_output_1$scaledResiduals, sites$fertilized)
```



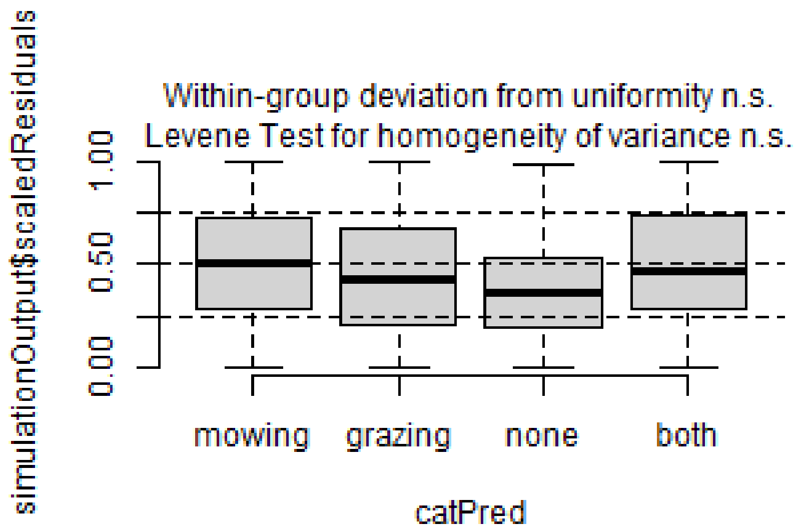

```
plotResiduals(simulation_output_2$scaledResiduals, sites$fertilized)
```




```r
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```




```r
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##                     GVIF Df GVIF^(1/(2*Df))
## esy4            3.487008  2        1.366511
## site.type       1.433677  2        1.094241
## eco.id          1.070092  2        1.017080
## obs.year        1.020821  1        1.010357
## esy4:site.type  4.545232  4        1.208355
```

```
car::vif(m_2)
```

```
##                      GVIF Df GVIF^(1/(2*Df))
## esy4            12.555214  2        1.882374
## site.type        1.596349  2        1.124040
## eco.id           1.775060  2        1.154259
## obs.year         1.032335  1        1.016039
## hydrology        1.341390  2        1.076190
## esy4:site.type   5.158284  4        1.227618
## esy4:eco.id      9.836240  3        1.463766
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##            R2m       R2c
## [1,] 0.1519338 0.6740039
MuMIn::r.squaredGLMM(m_2)
##            R2m       R2c
## [1,] 0.3118902 0.6957894
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df      AICc
## m_2 19 -1220.935
## m_1 14 -1190.718
```

Predicted values

Summary table

```
car::Anova(m_2, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##                   Chisq Df Pr(>Chisq)
## (Intercept)    454.2336  1  < 2.2e-16 ***
## esy4            12.9914  2    0.00151 **
## site.type        8.4332  2    0.01475 *
## eco.id           1.2761  2    0.52832
## obs.year         2.6442  1    0.10393
## hydrology       39.1188  2  3.202e-09 ***
## esy4:site.type   2.0074  4    0.73439
## esy4:eco.id      2.2107  3    0.52985
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m_2)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: y ~ esy4 * (site.type + eco.id) + obs.year + hydrology + (1 |
##     id.site)
##    Data: sites
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   -1222.2   -1138.2     630.1   -1260.2       597
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2199 -0.5504  0.0267  0.4701  5.5476
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  id.site  (Intercept) 0.005939 0.07706
##  Residual             0.004706 0.06860
## Number of obs: 616, groups:  id.site, 176
##
## Fixed effects:
##                        Estimate Std. Error t value
## (Intercept)            0.422513   0.019824  21.313
## esy4R22                0.030833   0.017862   1.726
## esy4R1A               -0.070045   0.023888  -2.932
```

```
## site.type.L          0.044359   0.017162   2.585
## site.type.Q          0.013490   0.012528   1.077
## eco.id664           -0.020491   0.018736  -1.094
## eco.id686           -0.014544   0.018088  -0.804
## obs.year2023        -0.021419   0.013172  -1.626
## hydrologyfresh       0.094613   0.017296   5.470
## hydrologymoist       0.113187   0.019863   5.698
## esy4R22:site.type.L -0.011640   0.022400  -0.520
## esy4R1A:site.type.L  0.034487   0.040824   0.845
## esy4R22:site.type.Q -0.015240   0.015333  -0.994
## esy4R1A:site.type.Q  0.001186   0.028152   0.042
## esy4R22:eco.id664   -0.027419   0.022264  -1.232
## esy4R22:eco.id686   -0.021702   0.020536  -1.057
## esy4R1A:eco.id686    0.016524   0.031184   0.530
```

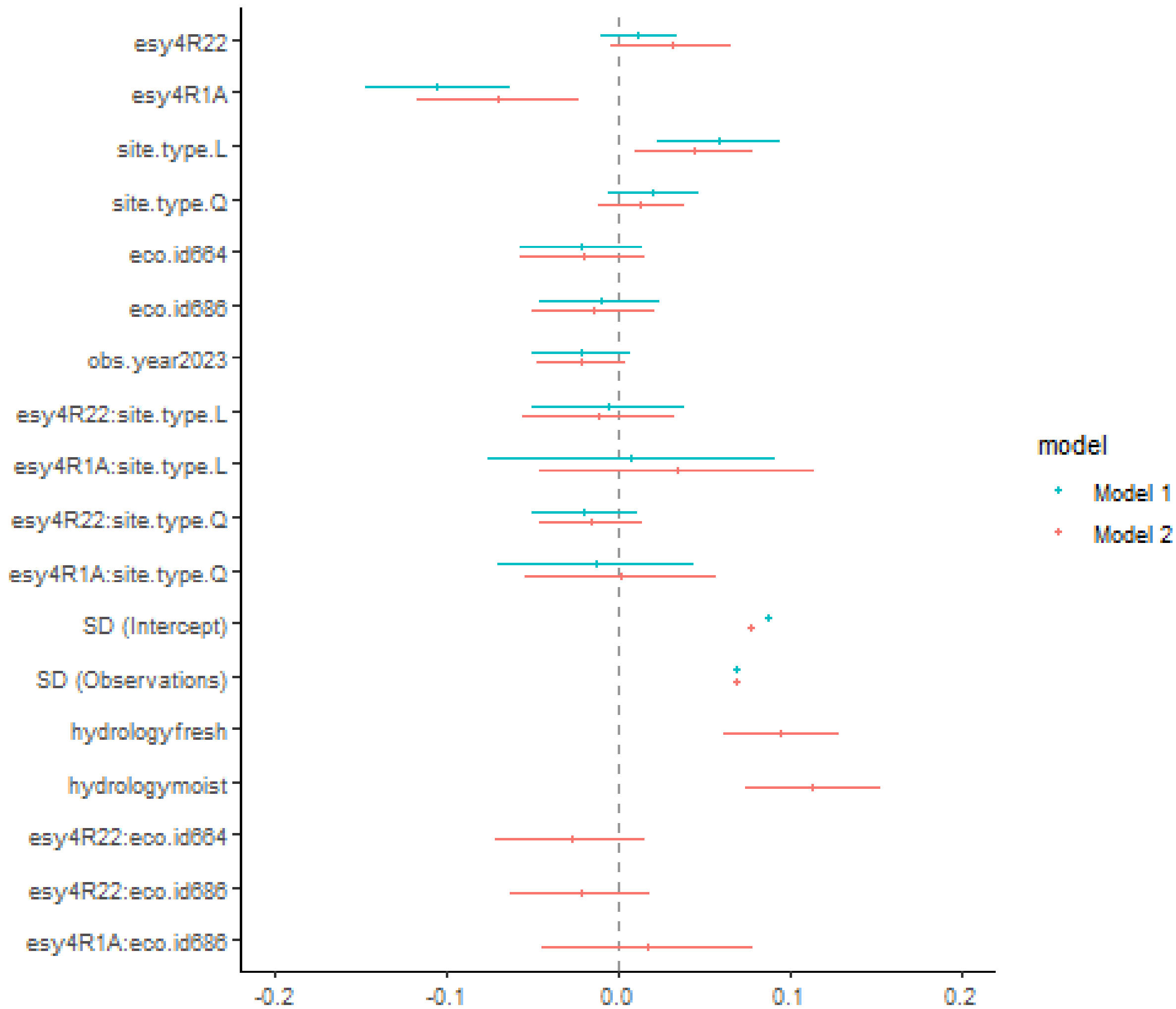
```
##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

Forest plot

```r
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  xlim(-0.2, 0.2) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_2,
  revpairwise ~ esy4,
  type = "response"
  ))
```

ESY EUNIS Habitat type

```
## $emmeans
##  esy4 emmean      SE  df lower.CL upper.CL
##  R     0.469 0.00918 157    0.451    0.488
##  R22   0.484 0.01240 299    0.459    0.508
##  R1A  nonEst      NA  NA       NA       NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
```

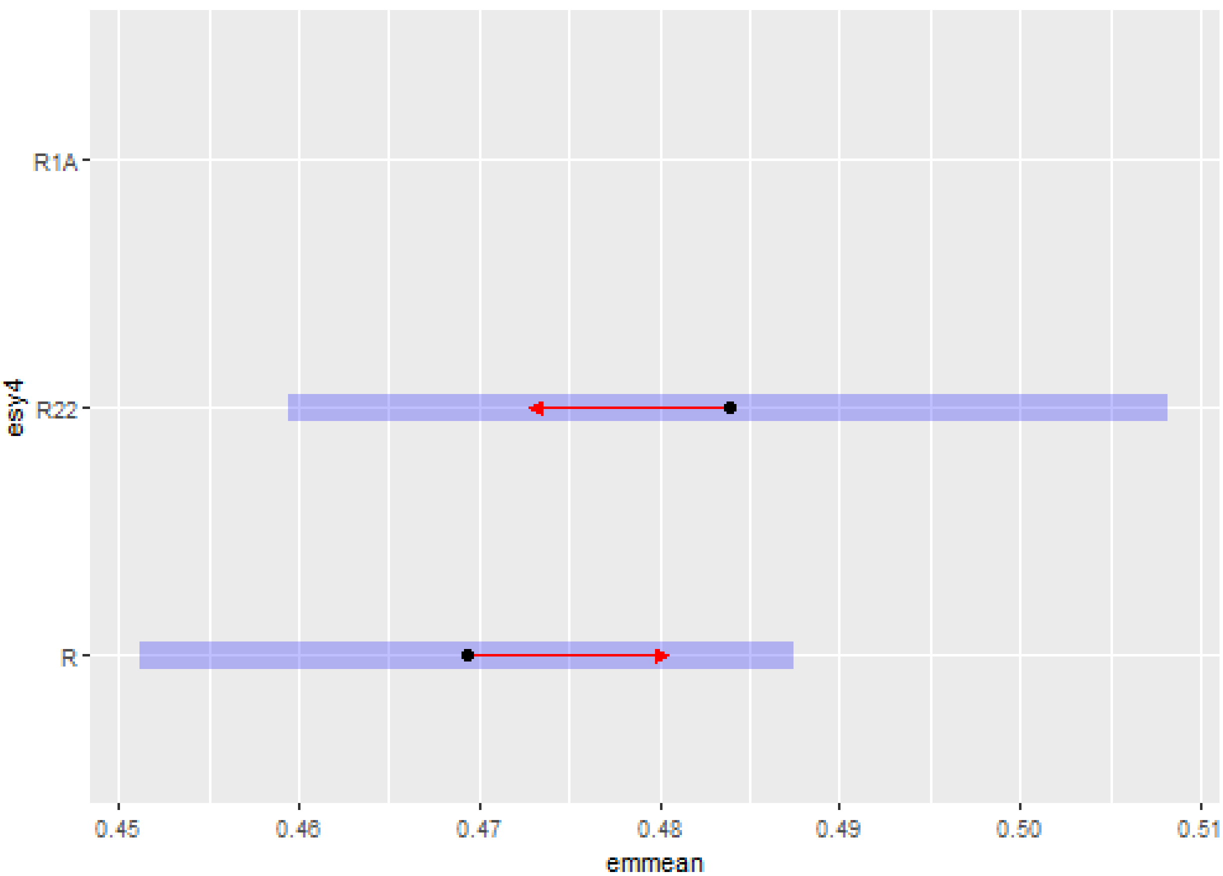
```
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##  contrast  estimate     SE  df t.ratio p.value
##  R22 - R     0.0145 0.0113 580   1.275  0.2028
##  R1A - R     nonEst     NA  NA      NA      NA
##  R1A - R22   nonEst     NA  NA      NA      NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 2 estimates
```

```r
plot(emm, comparison = TRUE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
(emm <- emmeans(
  m_2,
```

```r
  revpairwise ~ eco.id | esy4,
  type = "response"
  ))
```

Habiat type x Region

```
## $emmeans
## esy4 = R:
##  eco.id emmean     SE  df lower.CL upper.CL
##  654     0.481 0.0144 204    0.453    0.510
##  664     0.461 0.0144 219    0.432    0.489
##  686     0.467 0.0139 185    0.439    0.494
##
## esy4 = R22:
##  eco.id emmean     SE  df lower.CL upper.CL
##  654     0.512 0.0194 376    0.474    0.550
##  664     0.464 0.0166 311    0.431    0.497
##  686     0.476 0.0170 310    0.442    0.509
##
## esy4 = R1A:
##  eco.id emmean     SE  df lower.CL upper.CL
##  654     0.411 0.0229 194    0.366    0.456
##  664    nonEst     NA  NA       NA       NA
##  686     0.413 0.0298 414    0.354    0.472
##
## Results are averaged over the levels of: site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## esy4 = R:
##  contrast              estimate     SE  df t.ratio p.value
##  eco.id664 - eco.id654 -0.02049 0.0193 236  -1.063  0.5378
##  eco.id686 - eco.id654 -0.01454 0.0186 257  -0.782  0.7146
##  eco.id686 - eco.id664  0.00595 0.0187 251   0.318  0.9458
##
## esy4 = R22:
##  contrast              estimate     SE  df t.ratio p.value
##  eco.id664 - eco.id654 -0.04791 0.0226 387  -2.118  0.0876
##  eco.id686 - eco.id654 -0.03625 0.0224 408  -1.619  0.2386
##  eco.id686 - eco.id664  0.01166 0.0207 335   0.565  0.8390
##
## esy4 = R1A:
##  contrast              estimate     SE  df t.ratio p.value
##  eco.id664 - eco.id654   nonEst     NA  NA      NA      NA
##  eco.id686 - eco.id654  0.00198 0.0310 413   0.064  0.9491
##  eco.id686 - eco.id664   nonEst     NA  NA      NA      NA
##
## Results are averaged over the levels of: site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for varying family sizes
```

```r
plot(emm, comparison = TRUE)
```

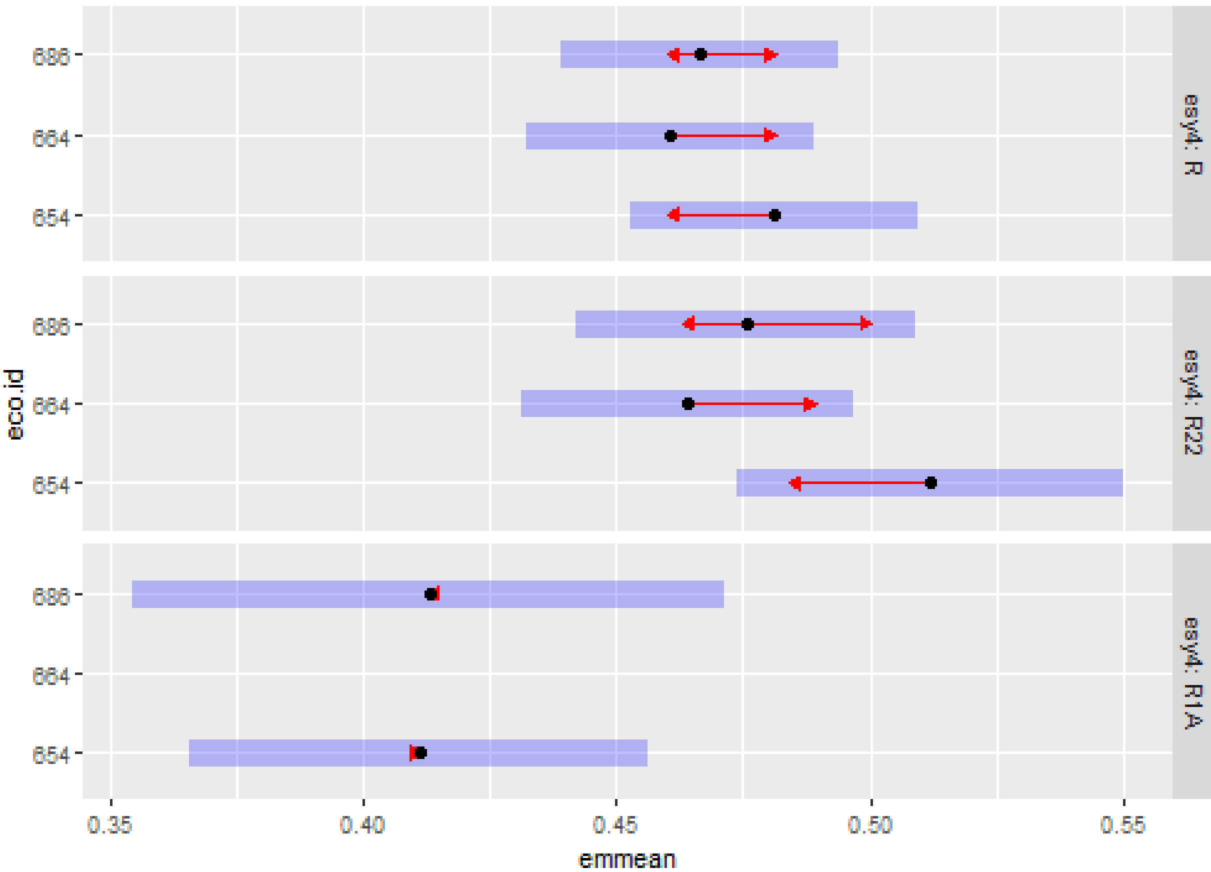
```
## Warning: Comparison discrepancy in group "R1A", eco.id654 - eco.id686:
```

```
##     Target overlap = 0.9675, overlap on graph = -0.7438

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
(emm <- emmeans(
  m_2,
  revpairwise ~ site.type | esy4,
  type = "response"
  ))
```

Habiat type x Site type

```
## $emmeans
## esy4 = R:
##  site.type emmean     SE  df lower.CL upper.CL
##  positive   0.444 0.0184 262    0.407    0.480
##  restored   0.458 0.0102 173    0.438    0.479
##  negative   0.506 0.0170 127    0.473    0.540
##
```

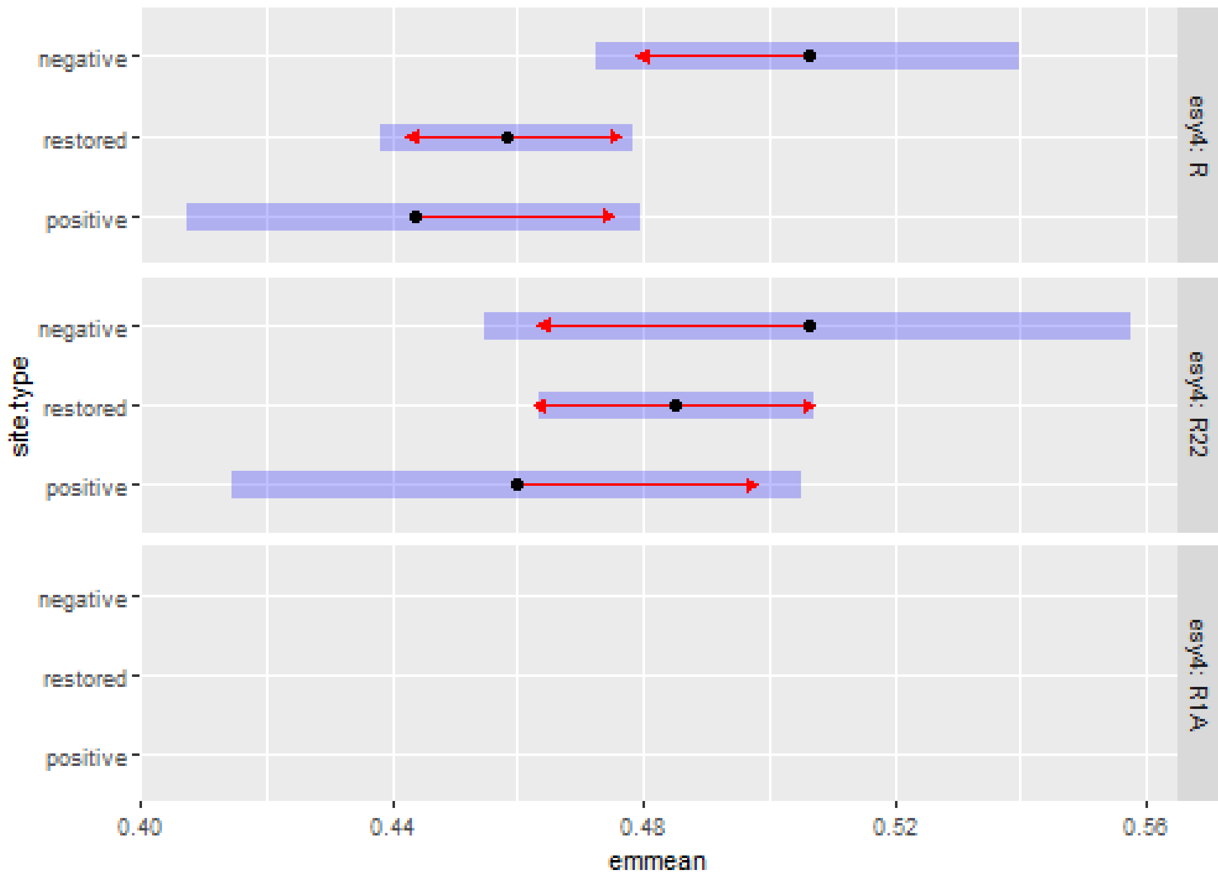
```
## esy4 = R22:
##  site.type emmean     SE  df lower.CL upper.CL
##  positive   0.460 0.0230 449    0.415    0.505
##  restored   0.485 0.0110 143    0.464    0.507
##  negative   0.506 0.0261 381    0.455    0.558
##
## esy4 = R1A:
##  site.type emmean     SE  df lower.CL upper.CL
##  positive  nonEst     NA  NA       NA       NA
##  restored  nonEst     NA  NA       NA       NA
##  negative  nonEst     NA  NA       NA       NA
##
## Results are averaged over the levels of: eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## esy4 = R:
##  contrast            estimate     SE  df t.ratio p.value
##  restored - positive   0.0148 0.0210 223   0.706  0.7599
##  negative - positive   0.0627 0.0250 177   2.509  0.0346
##  negative - restored   0.0479 0.0192 198   2.491  0.0360
##
## esy4 = R22:
##  contrast            estimate     SE  df t.ratio p.value
##  restored - positive   0.0253 0.0254 339   0.996  0.5797
##  negative - positive   0.0463 0.0348 417   1.331  0.3786
##  negative - restored   0.0210 0.0277 472   0.758  0.7288
##
## esy4 = R1A:
##  contrast            estimate     SE  df t.ratio p.value
##  restored - positive   0.0378 0.0340 446   1.113  0.5068
##  negative - positive   0.1115 0.0579 390   1.925  0.1329
##  negative - restored   0.0737 0.0538 375   1.371  0.3574
##
## Results are averaged over the levels of: eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
plot(emm, comparison = TRUE)
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     patchwork_1.3.0  ggbeeswarm_0.7.2
##  [5] lubridate_1.9.4  forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4
##  [9] purrr_1.0.4      readr_2.1.5      tidyr_1.3.1      tibble_3.2.1
```

```
## [13] ggplot2_3.5.2    tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.1.6            magrittr_2.0.3         multcomp_1.4-28
##  [7] compiler_4.5.1         mgcv_1.9-3             vctrs_0.6.5
## [10] pkgconfig_2.0.3        crayon_1.5.3           fastmap_1.2.0
## [13] backports_1.5.0        labeling_0.4.3         utf8_1.2.5
## [16] ggstance_0.3.7         promises_1.3.2         rmarkdown_2.29
## [19] tzdb_0.5.0             nloptr_2.2.1           bit_4.6.0
## [22] xfun_0.52              later_1.4.2            broom_1.0.8
## [25] parallel_4.5.1         R6_2.6.1               gap.datasets_0.0.6
## [28] stringi_1.8.7          qgam_2.0.0             RColorBrewer_1.1-3
## [31] car_3.1-3              boot_1.3-31            estimability_1.5.1
## [34] Rcpp_1.0.14            iterators_1.0.14       knitr_1.50
## [37] zoo_1.8-14             parameters_0.25.0      httpuv_1.6.16
## [40] Matrix_1.7-3           splines_4.5.1          timechange_0.3.0
## [43] tidyselect_1.2.1       rstudioapi_0.17.1      abind_1.4-8
## [46] yaml_2.3.10            MuMIn_1.48.11          doParallel_1.0.17
## [49] codetools_0.2-20       lattice_0.22-7         plyr_1.8.9
## [52] bayestestR_0.15.3      shiny_1.10.0           withr_3.0.2
## [55] evaluate_1.0.3         marginaleffects_0.25.1 survival_3.8-3
## [58] pillar_1.10.2          gap_1.6                carData_3.0-5
## [61] foreach_1.5.2          stats4_4.5.1           reformulas_0.4.1
## [64] insight_1.2.0          generics_0.1.4         vroom_1.6.5
## [67] rprojroot_2.0.4        hms_1.1.3              scales_1.4.0
## [70] minqa_1.2.8            xtable_1.8-4           glue_1.8.0
## [73] tools_4.5.1            data.table_1.17.2      lme4_1.1-37
## [76] mvtnorm_1.3-3          grid_4.5.1             rbibutils_2.3
## [79] datawizard_1.1.0       nlme_3.1-168           Rmisc_1.5.1
## [82] performance_0.13.0     beeswarm_0.4.0         vipor_0.4.7
## [85] Formula_1.2-5          cli_3.6.5              gtable_0.3.6
## [88] digest_0.6.37          pbkrtest_0.5.4         TH.data_1.1-3
## [91] farver_2.1.2           htmltools_0.5.8.1      lifecycle_1.0.4
## [94] mime_0.13              bit64_4.6.0-1          dotwhisker_0.8.4
## [97] MASS_7.3-65
```